SANDEEP SINGH

PhD — Mathematics & Data Science | Machine Learning, Applied Data Science

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PROFESSIONAL SUMMARY

PhD candidate in Mathematics & Data Science at Chandigarh University, currently in the third semester of doctoral research, building large-scale machine learning and deep learning systems for applied data science projects. Author of research manuscripts at the under-review, under-revision, and in-preparation stages targeting Springer Nature and Scientific Reports (Q1/Q2) journals, spanning air quality nowcasting, biomedical diagnostics, and cybersecurity detection. One year of experience teaching Applied Mathematics to B.E. Computer Science & Engineering students, including lecture planning, tutorial delivery, assessment design, and examination evaluation. Seeking a role to contribute to research at the intersection of machine learning, data science, and applied mathematics, combining hands-on research development with teaching and lab support.

EDUCATION

PhD — Mathematics & Data Science | Chandigarh University

2025 – Present (Currently 3rd Semester) | Mohali, Punjab | Supervisor: Prof. Ashok Pal

- Doctoral thesis: "Airshed Intelligence: Five-State Spatio-Temporal ML+DL and Policy Optimization" — air quality forecasting and control-policy modeling across Punjab, Haryana, Himachal Pradesh, Uttar Pradesh and Delhi.
- Focus areas: spatio-temporal machine learning, deep learning, statistical modeling, environmental data science, biostatistics, policy optimization.
- Targeting 3–4 publications in Springer Nature Q1/Q2 journals as part of doctoral output.

M.Sc. — Data Science | Chandigarh University

Completed June 2025 | GPA: 7.4 / 10

- Advanced coursework in machine learning, deep learning, health data analytics, and statistical inference.
- Capstone project: Developed a predictive modeling solution addressing real-world business and health data challenges.
- Presented research on ML techniques at the university's annual Data Science Symposium.

B.Sc. — Information Technology | Maharaja Ranjit Singh Punjab Technical University

Completed December 2019 | GPA: 7.09 / 9

- Core coursework: Python, Java, C++, Data Management, Cybersecurity.

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Research Scholar & Mathematics Faculty | Chandigarh University

2025 – Present | Mohali, Punjab

- Conducting doctoral research building spatio-temporal ML/DL forecasting systems across a five-state, 115-station, ~20.1-million-record air quality dataset (2021–2025).
- Taught Applied Mathematics to B.E. Computer Science & Engineering students (1 year) — planned lectures, conducted tutorials, developed assessments, and evaluated examinations.
- Designing and validating ML/DL forecasting and diagnostic models; preparing manuscripts for peer-reviewed Q1/Q2 journals.
- Built two reusable AI-assisted research tools: a structured three-pass manuscript peer-review pipeline (editorial desk-reject check, methodology critique, and response-letter drafting) grounded in COPE, ICMJE, and Springer publishing guidelines, and a scientific writing tool for manuscript and report drafting.
- Performing literature reviews, large-scale data engineering, and quantitative analysis aligned with doctoral research objectives.

Business Analyst Intern | KultureHire

Sep 2024 – Nov 2024

- Analyzed large datasets to identify trends, supporting strategic and operational decision-making.
- Developed MIS dashboards and visualizations using Tableau and Excel for stakeholder presentations.
- Collaborated cross-functionally to gather data requirements and align analytical outputs with project goals.

Data Analyst Intern | KultureHire

Jun 2024 – Aug 2024

- Translated client business requirements into actionable data insights for development and strategy teams.

- Conducted market research using SQL and Python; segmented customer demographics to support strategic recommendations.
- Delivered data-driven dashboards summarising findings for senior stakeholders.

Operations Manager | MS Royal Recency (Luxury Resort)
Nov 2021 – Feb 2023

- Managed daily operations, overseeing 13+ staff and executing data-driven marketing strategies that increased occupancy by 15%.

RESEARCH & TECHNICAL PROJECTS

Airshed Intelligence: Five-State Spatio-Temporal ML+DL and Policy Optimization [IN PREPARATION — DOCTORAL THESIS PROJECT]
Machine Learning | Deep Learning | Spatio-Temporal Forecasting | Environmental Policy | Large-Scale Data Engineering | Python

- Building a five-state air quality forecasting and policy-optimization framework covering Punjab, Haryana, Himachal Pradesh, Uttar Pradesh and Delhi, integrating 115 CPCB monitoring stations and approximately 20.1 million 15-minute-resolution records (2021–2025).
- Engineered a production-grade data infrastructure (station registry, standardized constants module, unified data loader) to harmonize heterogeneous multi-state, multi-agency air quality data.
- Identified and corrected critical data-quality issues invisible in raw records: a NOx unit inconsistency requiring a ×2.05 conversion factor, a network-wide Haryana 2025 sensor outage, DPCC Delhi stations reporting at effective 1-hour (not 15-minute) resolution, and inverted file-numbering conventions across agencies.
- Debugged and corrected a multi-city, multi-horizon Punjab forecasting pipeline (LSTM, GRU, Bi-LSTM), identifying and fixing a missing validation-data argument that had silently disabled all deep learning training callbacks (Early Stopping, Learning Rate Reduction, Model Checkpointing) across every training run.
- Targeting 3–4 publications in Springer Nature Q1/Q2 journals as part of doctoral output.

Hourly PM2.5 and PM10 Nowcasting Across 24 Haryana Cities [IN PREPARATION — MANUSCRIPT FINALISED FOR SUBMISSION]
Machine Learning | Deep Learning | Time-Series Forecasting | Statistical Testing | Springer Nature LaTeX

- Developed an hourly PM2.5/PM10 nowcasting system across 24 Haryana cities using six ML/DL models trained on CPCB/HSPCB monitoring data, including a fourth deep learning architecture (Bi-GRU) added to strengthen model comparison.
- Identified and fixed a hyperparameter-sharing bug across deep learning architectures that had been silently degrading model-to-model comparability.
- Added a persistence baseline with statistical significance testing to substantiate model performance claims against a naive benchmark.
- Took the manuscript through a structured three-pass peer-review pipeline (desk-reject check, methodology critique, response drafting), correcting bibliography metadata and recasting process-narrative prose into standard scientific reporting register.
- Prepared as a Springer Nature (sn-jnl) LaTeX manuscript targeting a Q1/Q2 journal.

Glioma Grade Prediction — ML Pipeline with SHAP & Clinical Threshold Optimization [UNDER REVISION]
Machine Learning | Ensemble Methods | SHAP Explainability | Neuro-Oncology | Python | Springer Journal Submission

- Built an end-to-end ML diagnostic pipeline on the UCI Glioma Grading Dataset (n = 839 patients, 20 somatic mutation markers + 3 clinical variables) to classify Low-Grade Glioma vs. Glioblastoma Multiforme (GBM).
- Implemented SMOTE strictly within the training partition to eliminate data leakage; tuned four classifiers (Logistic Regression, Random Forest, SVM, XGBoost) via RandomizedSearchCV plus Voting and Stacking ensembles.
- Evaluated models across ten metrics (ROC-AUC 0.926, MCC 0.738, Brier 0.102); used SHAP TreeExplainer to confirm IDH1 ($\bar{\varphi} = 0.207$) and Age as dominant features, aligning with WHO 2021 CNS classification.
- Designed a clinical cost-matrix threshold optimization framework ($C_{mn} = 10$, $C_{mp} = 2$) that shifted the deployment threshold from 0.5 to $\tau^* = 0.200$, achieving 97.1% GBM sensitivity and a 24% reduction in total clinical cost.
- Prepared a full peer-reviewed research paper (15 pages, 73 citations, svjour3 Springer format) covering rigorous methodology, statistical significance testing, biological validation, and clinical decision support recommendations.

Phishing & Android Malware Detection System [UNDER REVIEW]
Machine Learning | Heuristic Analysis | Model Validation | Adversarial Robustness Testing

- Built an automated detection system achieving 99.9% accuracy through advanced ML algorithms.
- Conducted extensive validation testing, including adversarial robustness checks; documented findings and presented technical reports.

AI Diagnostic Model — Bone Fracture & Gallstone Detection

Medical Imaging | Deep Learning | Computer Vision | Healthcare Analytics

- Led development of an AI diagnostic model across CT, X-ray, and ultrasound imaging modalities, achieving a 20% improvement in diagnostic accuracy.
- Collaborated with medical professionals and AI experts; validated models on 1,000+ clinical images, directly supporting evidence-based clinical decision-making.

Gen-Z Career Aspirations & Customer Demography Analysis

SQL | Python | Tableau | Statistical Analysis | Dashboard Reporting

- Designed and executed a full survey-to-dashboard analytics pipeline; extracted and analyzed data using SQL and Python.
- Produced MIS-style dashboards for stakeholder presentations, segmenting demographics and correlating findings with career preference data.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, XGBoost, SHAP, TensorFlow, Keras, PyTorch), R, SQL (MySQL, PostgreSQL)

Machine Learning & Deep Learning: LSTM, GRU, Bi-LSTM, Bi-GRU, Ensemble Methods (Voting, Stacking), Time-Series Forecasting, Computer Vision, NLP

Data Engineering: Large-scale multi-source data pipelines (20M+ records), data cleaning & standardization, multi-agency data harmonization, unit-consistency auditing

Statistics: Biostatistics, Statistical Modeling, Hypothesis Testing, Statistical Significance Testing, Regression Analysis, Survival Analysis

Explainability: SHAP TreeExplainer, LIME, Feature Importance, Decision Curve Analysis, Clinical Cost-Matrix Threshold Optimization

Academic Writing & Publishing: Springer Nature LaTeX (sn-jnl), peer-review response drafting, COPE/ICMJE guideline compliance, scientific reporting register

Data & MIS: Advanced Excel (Pivot Tables, Macros, Dashboards), Tableau, Power BI

Databases: MySQL, MongoDB, data extraction & query optimization

Teaching: Lecture planning, tutorial delivery, assessment design, examination evaluation

Soft Skills: Written & oral communication, presentation, cross-functional collaboration

CERTIFICATIONS

- How to Write and Publish a Scientific Paper (Project-Centered Course) — École Polytechnique, via Coursera — Feb 2026
- Mindshift: Break Through Obstacles to Learning and Discover Your Hidden Potential — McMaster University, via Coursera — Feb 2026
- Foundations of Teaching for Learning: Introduction — Commonwealth Education Trust, via Coursera — Jan 2026
- Numerical Methods for Engineers — The Hong Kong University of Science and Technology, via Coursera — Dec 2025
- University Teaching — The University of Hong Kong, via Coursera — Dec 2025
- Google Data Analyst Certificate — Coursera
- Data Mining, Data Analytics & Computer Vision — NPTEL
- Python, AI & IoT — Chandigarh University
- Excel, MySQL & Machine Learning — KultureHire

ADDITIONAL INFORMATION

Languages: English, Hindi, Punjabi